

- 2 A car travelling at a speed of 15.0 m s^{-1} can be brought to rest in 1.20 s when a uniform braking force is applied. The reaction time of the driver is 0.100 s .

What is the minimum distance at which the driver must notice a stationary object in order to avoid hitting it?

A 1.50 m

B 7.50 m

C 9.00 m

D 10.5 m

Ans: D

$$v = u + at \Rightarrow 0 = 15.0 + a(1.20) \Rightarrow a = -\frac{15.0}{1.20} \text{ m s}^{-2}$$

$$v^2 = u^2 + 2as \Rightarrow 0 = 15.0^2 + 2\left(-\frac{15.0}{1.20}\right)s \Rightarrow s = 9.00 \text{ m}$$

$$\text{Thinking distance} = 15.0 \text{ m s}^{-1} \times 0.100 \text{ s} = 1.50 \text{ m}$$

$$\text{Minimum distance} = 1.50 + 9.00 = 10.5 \text{ m}$$

- 3 Ball P is projected upwards from the bottom of a smooth inclined plane with an initial speed that